

BILKENT UNIVERSITY
ELECTRICAL AND ELECTRONICS ENGINEERING
MATH 241
ENGINEERING MATHEMATICS I
FINAL EXAM **SOLUTION**
8-01-2013

Instructions:

- There are ten questions in the exam. They have different weights. Solve all.
 - Duration of the exam is 180 minutes.
 - The exam is closed book, closed notes.
 - No calculators are allowed.
 - Please read each question carefully, explain your answers clearly, and show all your work to receive full credit.
 - Boldface capital letters are matrices and boldface small letters are vectors.
 - Do not detach pages.
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Surname:.....

Name:.....

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Section:.....

Signature:.....

Question	Grade
1	
2	
3	
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9	
10	
Total:	

Q1. (10 points) Let $\mathbf{A}_{m \times n}$ be a matrix for which $\mathbf{Ax} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ has no solution, and $\mathbf{Ax} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ has

exactly one solution.

- (a) Give all possible information about m, n , and the rank of $\mathbf{A}$.
- (b) Find all solutions to $\mathbf{Ax} = \mathbf{0}$.
- (c) Write down an example of a matrix $\mathbf{A}$ that fits the description in part (a).

Solution:

- (a) Since the right hand sides belong to $\mathbb{R}^3$, $m = 3$.

$$\mathbf{Ax} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \text{ has no solution means } r(\mathbf{A}) < m \text{ (i.e. } \text{Col}(\mathbf{A}) \neq \mathbb{R}^3)$$

$$\mathbf{Ax} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \text{ has exactly one solution means } r(\mathbf{A}) = n \text{ (full rank)}$$

Therefore $n < m$ which means there are two possibilities $m = 3, n = 2$ or $m = 3, n = 1$.

$$(b) \mathbf{Ax} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \text{ has exactly one solution means } \mathbf{Ax} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ also has a unique solution which}$$

is either $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ or $[0]$.

$$(c) \mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \text{ or } \mathbf{A} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

Q2. (6 points) For a homogeneous constant coefficient difference equation, the roots of its characteristic equation are given as

$$r_1 = 2 \text{ with multiplicity } 2$$

$$r_2 = 3 \text{ with multiplicity } 2$$

$$r_3 = -1$$

(a) Obtain the general solution

(b) Find a difference equation that accepts the above solutions as its general solution

Solution:

(a) $y_n = \alpha_1 2^n + \alpha_2 n 2^n + \alpha_3 3^n + \alpha_4 n 3^n + \alpha_5 (-1)^n$

(b) The characteristic polynomial is

$$p(r) = (r-2)^2(r-3)^2(r+1) = (r^2 - 4r + 4)(r^2 - 6r + 9)(r+1)$$

$$= (r^4 - 6r^3 + 9r^2 - 4r^3 + 24r^2 - 36r + 4r^2 - 24r + 36)(r+1)$$

$$= (r^4 - 10r^3 + 37r^2 - 60r + 36)(r+1)$$

$$= r^5 - 9r^4 + 27r^3 - 23r^2 - 24r + 36$$

$$\text{Therefore } y_n = 9y_{n-1} - 27y_{n-2} + 23y_{n-3} + 24y_{n-4} - 36y_{n-5}$$

Q3. (10 points) Let $\mathbf{A}_{n \times n}$ be a real matrix and $\mathbf{x} \in \mathbb{R}^n$ is a non-zero vector (i.e., $\mathbf{x} \neq \mathbf{0}$). It is given that $\mathbf{x} \notin \text{Null}(\mathbf{A})$, $\mathbf{x} \notin \text{Null}(\mathbf{A}^2)$, ..., $\mathbf{x} \notin \text{Null}(\mathbf{A}^{k-1})$ but $\mathbf{x} \in \text{Null}(\mathbf{A}^k)$, where $\text{Null}(\cdot)$ denotes the null space of $(\cdot)$. Prove that the set $\{\mathbf{x}, \mathbf{A}\mathbf{x}, \mathbf{A}^2\mathbf{x}, \dots, \mathbf{A}^{k-1}\mathbf{x}\}$ is linearly independent.

Solution:

$$\alpha_1 \mathbf{x} + \alpha_2 \mathbf{A}\mathbf{x} + \alpha_3 \mathbf{A}^2\mathbf{x} + \dots + \alpha_k \mathbf{A}^{k-1}\mathbf{x} = \mathbf{0}$$

Multiplying by $\mathbf{A}^{k-1}$ we obtain

$$\alpha_1 \mathbf{A}^{k-1}\mathbf{x} + \alpha_2 \mathbf{A}^k\mathbf{x} + \alpha_3 \mathbf{A}^{k+1}\mathbf{x} + \dots + \alpha_k \mathbf{A}^{2k-2}\mathbf{x} = \mathbf{0}$$

Since $\mathbf{A}^{k+i}\mathbf{x} = \mathbf{0}$ $i = 0, 1, \dots$, we end up with $\alpha_1 \mathbf{A}^{k-1}\mathbf{x} = \mathbf{0}$. This means that $\alpha_1 = 0$ because $\mathbf{A}^{k-1}\mathbf{x} \neq \mathbf{0}$.

Now consider multiplying by $\mathbf{A}^{k-2}$

$\alpha_1 \mathbf{A}^{k-2}\mathbf{x} + \alpha_2 \mathbf{A}^{k-1}\mathbf{x} + \alpha_3 \mathbf{A}^k\mathbf{x} + \dots + \alpha_k \mathbf{A}^{2k-3}\mathbf{x} = \mathbf{0}$ which means $\alpha_1 \mathbf{A}^{k-2}\mathbf{x} + \alpha_2 \mathbf{A}^{k-1}\mathbf{x} = \mathbf{0}$. Since we already know that $\alpha_1 = 0$, we have $\alpha_2 \mathbf{A}^{k-1}\mathbf{x} = \mathbf{0}$ and therefore by the same reasoning $\alpha_2 = 0$.

Continuing in this manner we can show that all alphas are zero, and therefore the given set is LI.

Q4. (15 points) It is given that $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is an orthonormal basis for $\mathbb{R}^n$ and the matrix $\mathbf{R}_k$ is given by $\mathbf{R}_k = \mathbf{I} - 2\mathbf{e}_k\mathbf{e}_k^T$ $1 \leq k \leq n$

(a) Show that $\mathbf{R}_k\mathbf{R}_l = \mathbf{R}_l\mathbf{R}_k$

(b) Show that $(\mathbf{R}_k)^2 = \mathbf{I}$

(c) What are the eigenvalues and the corresponding eigenvectors of $\mathbf{R}_k$?

Solution:

(a) If $k = l$ we have the equality satisfied trivially. If $k \neq l$

$$\mathbf{R}_k\mathbf{R}_l = (\mathbf{I} - 2\mathbf{e}_k\mathbf{e}_k^T)(\mathbf{I} - 2\mathbf{e}_l\mathbf{e}_l^T) = \mathbf{I} - 2\mathbf{e}_k\mathbf{e}_k^T - 2\mathbf{e}_l\mathbf{e}_l^T + 4\mathbf{e}_k\mathbf{e}_k^T\mathbf{e}_l\mathbf{e}_l^T = \mathbf{I} - 2\mathbf{e}_k\mathbf{e}_k^T - 2\mathbf{e}_l\mathbf{e}_l^T$$

$$\mathbf{R}_l\mathbf{R}_k = (\mathbf{I} - 2\mathbf{e}_l\mathbf{e}_l^T)(\mathbf{I} - 2\mathbf{e}_k\mathbf{e}_k^T) = \mathbf{I} - 2\mathbf{e}_l\mathbf{e}_l^T - 2\mathbf{e}_k\mathbf{e}_k^T + 4\mathbf{e}_l\mathbf{e}_l^T\mathbf{e}_k\mathbf{e}_k^T = \mathbf{I} - 2\mathbf{e}_l\mathbf{e}_l^T - 2\mathbf{e}_k\mathbf{e}_k^T$$

because $\mathbf{e}_k^T\mathbf{e}_l = \mathbf{e}_l^T\mathbf{e}_k = \delta_{lk} = 0$ if $l \neq k$. Thus $\mathbf{R}_k\mathbf{R}_l = \mathbf{R}_l\mathbf{R}_k$.

$$(b) (\mathbf{R}_k)^2 = (\mathbf{I} - 2\mathbf{e}_k\mathbf{e}_k^T)(\mathbf{I} - 2\mathbf{e}_k\mathbf{e}_k^T) = \mathbf{I} - 4\mathbf{e}_k\mathbf{e}_k^T + 4\mathbf{e}_k\mathbf{e}_k^T\mathbf{e}_k\mathbf{e}_k^T = \mathbf{I} - 4\mathbf{e}_k\mathbf{e}_k^T + 4\mathbf{e}_k\mathbf{e}_k^T = \mathbf{I}$$

(c)

$$(\mathbf{I} - 2\mathbf{e}_k\mathbf{e}_k^T)\mathbf{x} = \lambda\mathbf{x} \Rightarrow (\mathbf{I} - 2\mathbf{e}_k\mathbf{e}_k^T)(\alpha_1\mathbf{e}_1 + \alpha_2\mathbf{e}_2 + \dots + \alpha_n\mathbf{e}_n) = \lambda(\alpha_1\mathbf{e}_1 + \alpha_2\mathbf{e}_2 + \dots + \alpha_n\mathbf{e}_n)$$

$$\Rightarrow \alpha_1\mathbf{e}_1 + \alpha_2\mathbf{e}_2 + \dots + \alpha_n\mathbf{e}_n - 2\alpha_k\mathbf{e}_k = \lambda(\alpha_1\mathbf{e}_1 + \alpha_2\mathbf{e}_2 + \dots + \alpha_n\mathbf{e}_n)$$

$$\Rightarrow \alpha_1(1-\lambda)\mathbf{e}_1 + \alpha_2(1-\lambda)\mathbf{e}_2 + \dots + \alpha_k(1-2-\lambda)\mathbf{e}_k + \dots + \alpha_n(1-\lambda)\mathbf{e}_n = \mathbf{0}$$

Since the vectors are LI, the coefficients must be zero and therefore there are two eigenvalues: $\lambda_1 = -1$ and $\lambda_2 = 1$.

Let us now find the eigenvectors

$$\lambda_1 = -1:$$

$$(\mathbf{I} - 2\mathbf{e}_k\mathbf{e}_k^T)\mathbf{x} = \lambda\mathbf{x} \Rightarrow (\mathbf{I} - 2\mathbf{e}_k\mathbf{e}_k^T)\mathbf{x} = -\mathbf{x} \Rightarrow -2\mathbf{e}_k\mathbf{e}_k^T\mathbf{x} = -2\mathbf{x} \Rightarrow \mathbf{e}_k\mathbf{e}_k^T\mathbf{x} = \mathbf{x}$$

$$\Rightarrow \mathbf{e}_k\mathbf{e}_k^T(\alpha_1\mathbf{e}_1 + \alpha_2\mathbf{e}_2 + \dots + \alpha_n\mathbf{e}_n) = \alpha_1\mathbf{e}_1 + \alpha_2\mathbf{e}_2 + \dots + \alpha_n\mathbf{e}_n$$

$$\Rightarrow \mathbf{e}_k\alpha_k = (\alpha_1\mathbf{e}_1 + \alpha_2\mathbf{e}_2 + \dots + \alpha_n\mathbf{e}_n) \Rightarrow \alpha_k = 1, \alpha_j = 0, j \neq k$$

$$\Rightarrow \mathbf{x} = \mathbf{e}_k$$

$$\lambda_1 = 1:$$

$$(\mathbf{I} - 2\mathbf{e}_k\mathbf{e}_k^T)\mathbf{x} = \lambda\mathbf{x} \Rightarrow (\mathbf{I} - 2\mathbf{e}_k\mathbf{e}_k^T)\mathbf{x} = \mathbf{x} \Rightarrow \mathbf{x} - 2\mathbf{e}_k\mathbf{e}_k^T\mathbf{x} = \mathbf{x} \Rightarrow \mathbf{x} - 2\mathbf{e}_k\mathbf{e}_k^T\mathbf{x} - \mathbf{x} = \mathbf{x} - \mathbf{x} = \mathbf{0}$$

$$\Rightarrow \mathbf{e}_k\mathbf{e}_k^T\mathbf{x} = \mathbf{0} \Rightarrow \mathbf{e}_k\mathbf{e}_k^T(\alpha_1\mathbf{e}_1 + \alpha_2\mathbf{e}_2 + \dots + \alpha_n\mathbf{e}_n) = \mathbf{0} \Rightarrow \mathbf{e}_k\mathbf{e}_k^T\alpha_k\mathbf{e}_k = \mathbf{0} \Rightarrow \mathbf{e}_k\alpha_k = \mathbf{0} \Rightarrow \alpha_k = 0$$

The other alphas may be anything. Thus any combination of the vectors

$\mathbf{e}_j$ $j = 1, \dots, n$ $j \neq k$ is an eigenvector. Thus $\mathbf{e}_j$ $j = 1, \dots, n$ $j \neq k$ can be taken as the

eigenvectors and therefore $\lambda_1 = 1$ has multiplicity $n-1$.

To check:

$$(\mathbf{I} - 2\mathbf{e}_k\mathbf{e}_k^T)\mathbf{x} = \lambda\mathbf{x} \Rightarrow \mathbf{e}_k - 2\mathbf{e}_k\mathbf{e}_k^T\mathbf{e}_k = \lambda\mathbf{e}_k \Rightarrow \mathbf{e}_k - 2\mathbf{e}_k = \lambda\mathbf{e}_k \Rightarrow -\mathbf{e}_k = \lambda\mathbf{e}_k \Rightarrow \lambda = -1$$

Thus $\lambda_1 = -1$ with the corresponding eigenvector $\mathbf{e}_k$

$$\text{Let us now try } \mathbf{e}_j \text{ } j \neq k: (\mathbf{I} - 2\mathbf{e}_k\mathbf{e}_k^T)\mathbf{x} = \lambda\mathbf{x} \Rightarrow \mathbf{e}_j - 2\mathbf{e}_k\mathbf{e}_k^T\mathbf{e}_j = \lambda\mathbf{e}_j \Rightarrow \mathbf{e}_j = \lambda\mathbf{e}_j \Rightarrow \lambda = 1$$

Thus $\lambda_2 = 1$ with multiplicity $n-1$ and with the corresponding eigenvectors

$$\mathbf{e}_j \text{ } j = 1, \dots, n, j \neq k$$

Q5. (12 points) Consider the matrix $\mathbf{A} = \begin{bmatrix} -1 & 3 & -1 & 1 \\ -3 & 5 & 1 & -1 \\ 10 & -10 & -10 & 14 \\ 4 & -4 & -4 & 8 \end{bmatrix}$.

It is known that $\begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ is an eigenvector of $\mathbf{A}$. It is further known that its determinant is zero.

Also one of the eigenvalues is known to be 4. Find all eigenvalues. **Do not attempt to find the roots of the char. poly..**

Solution:

$\lambda_1 = 4$ is given.

$\lambda_2 = 0$ since the determinant is 0.

$$\begin{bmatrix} -1 & 3 & -1 & 1 \\ -3 & 5 & 1 & -1 \\ 10 & -10 & -10 & 14 \\ 4 & -4 & -4 & 8 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 0 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \text{ and therefore } \lambda_3 = 0$$

Also $\text{Trace}(\mathbf{A}) = 2 = 4 + 0 + 2 + \lambda_4 \Rightarrow \lambda_4 = -4$

Q6. (10 points) Let $\mathbf{A}_{m \times n}$ and $\mathbf{B}_{n \times m}$ be two matrices such that $m > n$.

Prove that $\mathbf{AB}$ is not invertible.

Solution:

Since $\mathbf{B}$ has less number of rows than unknowns, $\mathbf{Bx} = \mathbf{0}$ has many solutions and hence at least one non-zero solution say $\mathbf{s} \neq \mathbf{0}$.

$$(\mathbf{AB})\mathbf{s} = \mathbf{A}(\mathbf{Bs}) = \mathbf{A}\mathbf{0} = \mathbf{0}$$

This means $(\mathbf{AB})\mathbf{x} = \mathbf{0}$ has a nonzero solution and therefore it has many solutions and therefore it is not invertible.

Q7. (8 points) Find the best-fit values of a and b so that $y = a + bx$ fits the data given in the table

x	1	2	3	4	5
y	12	9	6	2	1

Solution:

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} \approx \begin{bmatrix} 12 \\ 9 \\ 6 \\ 2 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} 12 \\ 9 \\ 6 \\ 2 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 5 & 15 \\ 15 & 55 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 30 \\ 61 \end{bmatrix}$$

$$\begin{bmatrix} a \\ b \end{bmatrix} = \frac{1}{50} \begin{bmatrix} 55 & -15 \\ -15 & 5 \end{bmatrix} \begin{bmatrix} 30 \\ 61 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 11 & -3 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 30 \\ 61 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 147 \\ -29 \end{bmatrix} = \begin{bmatrix} 14.7 \\ -2.9 \end{bmatrix}$$

Q8. (9 points) Consider the differential equation given by

$$\frac{d^2 y}{dt^2} + c_1 \frac{dy}{dt} + c_2 y = 0, \text{ where } c_1 \text{ and } c_2 \text{ are constants.}$$

- (a) If c_1 and c_2 are real, $c_1 > 0$, $c_2 < 0$, what can you say about the stability of the general solution of this differential equation?
- (b) If $c_2 = 0$ and c_1 is a complex number given as $c_1 = \alpha + j\beta$, what condition on c_1 (i.e., on α and β) will guarantee that every solution to this differential equation satisfies $\lim_{t \rightarrow \infty} y(t)$ is stable.
- (c) If $c_1 = 0$ and c_2 is real and positive, what can you say about the stability of the general solution of this differential equation?

Solution:

- (a) $\lambda^2 + c_1 \lambda + c_2 = 0 \Rightarrow \lambda = \frac{-c_1 \pm \sqrt{c_1^2 - 4c_2}}{2}$. This means that one root is positive.

Therefore the system is unstable. In fact the same conclusion can be arrived at by observing that the coefficients are of mixed sign.

- (b) $\lambda^2 + c_1 \lambda = 0 \Rightarrow \lambda = 0, -c_1 \Rightarrow \lambda = 0, -\alpha - j\beta$.

$$y(t) = A + be^{-\alpha t} e^{-j\beta t}$$

The system is unstable if $\alpha < 0$, and is stable if $\alpha \geq 0$.

- (c) $\lambda^2 + c_2 = 0 \Rightarrow \lambda = \pm j\sqrt{c_2}$.

$$y(t) = A \cos(\sqrt{c_2} t) + B \sin(\sqrt{c_2} t)$$

The solution is therefore oscillatory and stable.

Q9. (10 points) Solve the following initial value problem using the Laplace Transform.

$$y'' - 3y' + 2y = 3\cos t + \sin t \quad ; \quad y(0)=1, \quad y'(0)=1.$$

Solution:

$$\left[s^2 Y(s) - sy(0) - y'(0) \right] - 3[sY(s) - y(0)] + 2Y(s) = 3 \frac{s}{s^2+1} + \frac{1}{s^2+1}$$

$$(s^2 - 3s + 2)Y(s) = 3 \frac{s}{s^2+1} + \frac{1}{s^2+1} + s + 1 - 3 = \frac{3s+1}{s^2+1} + s - 2$$

$$Y(s) = \frac{3s+1}{(s^2+1)(s^2-3s+2)} + \frac{s-2}{s^2-3s+2} = \frac{3s+1}{(s^2+1)(s-2)(s-1)} + \frac{s-2}{(s-2)(s-1)}$$

$$= \frac{3s+1}{(s^2+1)(s-2)(s-1)} + \frac{1}{s-1}$$

$$\frac{3s+1}{(s^2+1)(s-2)(s-1)} = \frac{As+B}{s^2+1} + \frac{C}{s-2} + \frac{D}{s-1} = \frac{(As+B)(s-2)(s-1) + C(s^2+1)(s-1) + D(s^2+1)(s-2)}{(s^2+1)(s-2)(s-1)}$$

$$= \frac{As^3 - 3As^2 + 2As + Bs^2 - 3Bs + 2B + Cs^3 - Cs^2 + Cs - C + Ds^3 - 2Ds^2 + Ds - 2D}{(s^2+1)(s-2)(s-1)}$$

$$A + C + D = 0 \quad A = 3/5$$

$$-3A + B - C - 2D = 0 \quad \Rightarrow \quad B = -4/5$$

$$2A - 3B + C + D = 3 \quad C = 7/5$$

$$2B - C - 2D = 1 \quad D = -2$$

$$Y(s) = \frac{\frac{3}{5}s - \frac{4}{5}}{s^2+1} + \frac{\frac{7}{5}}{s-2} + \frac{-2}{s-1} + \frac{1}{s-1} = \frac{\frac{3}{5}s}{s^2+1} + \frac{-\frac{4}{5}}{s^2+1} + \frac{\frac{7}{5}}{s-2} + \frac{-1}{s-1}$$

$$y(t) = \frac{3}{5} \cos t - \frac{4}{5} \sin t + \frac{7}{5} e^{2t} - e^t$$

Q10. (10 points) The following linear system of differential equations are given:

$$\frac{dx_1}{dt} = -x_2(t)$$

$$\frac{dx_2}{dt} = x_1(t)$$

It is also given that $x_1(0) = 1$ and $x_2(0) = 1$.

(a) Find the state transition matrix.

(b) Find the solution of this system subject to the given initial conditions.

Solution:

$$(a) \frac{d}{dt} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}. \text{ Thus in } \mathbf{x}' = \mathbf{Ax}, \mathbf{A} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$\det \begin{bmatrix} -\lambda & -1 \\ 1 & -\lambda \end{bmatrix} = \lambda^2 + 1 = 0 \Rightarrow \lambda = \pm j$$

$$e^{At} = \alpha \mathbf{A} + \beta \mathbf{I}$$

$$e^{jt} = \alpha j + \beta$$

$$e^{-jt} = -\alpha j + \beta$$

$$\Rightarrow \alpha = \sin t$$

$$\beta = \cos t$$

$$e^{At} = \alpha \mathbf{A} + \beta \mathbf{I} = \sin t \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} + \cos t \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{bmatrix}$$

$$\text{The state transition matrix is } e^{\mathbf{A}(t-t_0)} = \begin{bmatrix} \cos(t-t_0) & -\sin(t-t_0) \\ \sin(t-t_0) & \cos(t-t_0) \end{bmatrix}$$

(b)

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = e^{\mathbf{A}t} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \cos t - \sin t \\ \sin t + \cos t \end{bmatrix}$$

Thus $x_1(t) = \cos t - \sin t$ and $x_2(t) = \sin t + \cos t$.